ASCE 41-23 Tier 1 Checklists

FIRM:	
PROJECT NAME:	
SEISMICITY LEVEL:	
PROJECT NUMBER:	
COMPLETED BY:	
DATE COMPLETED:	
REVIEWED BY:	
REVIEW DATE:	

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

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Project Name _____
Project Number _____

17.2CP Structural Checklist for Building Types W1: Wood Light Frames, Small Residential

Table 17-4. Collapse Prevention Structural Checklist for Building Types W1.

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Low and Moderate Seismicity							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REDUNDANCY: The number of lines of shear walls in each principal direction is greater than or equal to 2.	5.5.1.1	A.3.2.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	SHEAR STRESS CHECK: The shear stress in the shear walls, calculated using the Quick Check procedure of Section 4.4.3.3, is less than the following values: Structural panel sheathing 1,000 lb/ft (14.6 kN/m), Diagonal sheathing 700 lb/ft (10.2 kN/m), Straight sheathing 100 lb/ft (1.5 kN/m), and All other conditions 100 lb/ft (1.5 kN/m).	5.5.3.1.1	A.3.2.7.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	STUCCO (EXTERIOR PLASTER) SHEAR WALLS: Multistory buildings do not rely on exterior stucco walls as the primary seismic-force-resisting system.	5.5.3.6.1	A.3.2.7.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	GYPSUM WALLBOARD OR PLASTER SHEAR WALLS: Interior plaster or gypsum wallboard is not used for shear walls on buildings more than one story high with the exception of the uppermost level of a multistory building.	5.5.3.6.1	A.3.2.7.3	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	NARROW WOOD SHEAR WALLS: Narrow wood shear walls with an aspect ratio greater than 2-to-1 are not used to resist seismic forces.	5.5.3.6.1	A.3.2.7.4	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	WALLS CONNECTED THROUGH FLOORS: Shear walls have an interconnection between stories to transfer overturning and shear forces through the floor.	5.5.3.6.2	A.3.2.7.5	

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C ☐	**NC** ☐	**N/A** ☐	**U** ☐	HILLSIDE SITE: For structures that are taller on at least one side by more than one-half story because of a sloping site, all shear walls on the downhill slope have an aspect ratio less than 1-to-1.	5.5.3.6.3	A.3.2.7.6
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CRIPPLE WALLS: Cripple walls below first-floor-level shear walls are braced to the foundation with wood structural panels.	5.5.3.6.4	A.3.2.7.7
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	OPENINGS: Walls with openings greater than 80% of the length are braced with wood structural panel shear walls with aspect ratios of not more than 1.5-to-1 or are supported by adjacent construction through positive ties capable of transferring the seismic forces.	5.5.3.6.5	A.3.2.7.8
Connections						
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	WOOD POSTS: There is a positive connection of wood posts to the foundation.	5.7.3.3	A.5.3.3
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	WOOD SILLS: All wood sills are bolted to the foundation.	5.7.3.3	A.5.3.4
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	GIRDER–COLUMN CONNECTION: There is a positive connection using plates, connection hardware, or straps between the girder and the column support.	5.7.4.1	A.5.4.1

Status	**Evaluation Statement**	**Tier 2 Reference**	**Commentary Reference**	**Comments**
High Seismicity (Complete the Following Items in Addition to the Items for Low and Moderate Seismicity)				
Connections				
C ☐ **NC** ☐ **N/A** ☐ **U** ☐	WOOD SILL BOLTS: Sill bolts are spaced at 6 ft (1.8 m) or less with acceptable edge and end distance provided for wood and concrete.	5.7.3.3	A.5.3.7	
Diaphragms				
C ☐ **NC** ☐ **N/A** ☐ **U** ☐	DIAPHRAGM CONTINUITY: Floor and roof diaphragms do not have expansion joints or vertical offsets, such as split levels, sawtooth, or clerestory configurations.	5.6.1.1	A.4.1.1	

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C ☐	**NC** ☐	**N/A** ☐	**U** ☐	ROOF CHORD CONTINUITY: All chord elements are continuous, regardless of changes in roof elevation.	5.6.1.1	A.4.1.3
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	STRAIGHT SHEATHING: All straight-sheathed diaphragms have horizontal spans less than 24 ft (7.3 m) and aspect ratios less than 2-to-1 in the direction being considered.	5.6.2	A.4.2.1
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	DIAGONALLY SHEATHED AND UNBLOCKED DIAPHRAGMS: All diagonally sheathed or unblocked wood structural panel diaphragms have horizontal spans less than 40 ft (12.2 m) and have aspect ratios less than or equal to 4-to-1.	5.6.2	A.4.2.2
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	OTHER DIAPHRAGMS: The diaphragms do not consist of a system other than wood, steel deck, concrete, or horizontal bracing.	5.6.5	A.4.7.1

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